

# SS2P2 — Formulation

Stable State-Space Point Process (*as implemented*)

SS2P2 keeps S2P2’s expressive latent-linear-Hawkes (LLH) backbone and replaces its two output heads with a **gated-bounded (G1) rate** and a **rate-neutral mark head**:

$$\mathbf{SS2P2} = \underbrace{\text{LLH backbone} \rightarrow u(t)}_{\text{unchanged}} \implies \underbrace{\lambda(t) = s \text{softplus}(w_{\text{eff}}^\top h(u) + b)}_{\text{G1 bounded rate}} \times \underbrace{p^*(k | t) = \text{softmax}(\text{MLP}(u))_k}_{\text{decoupled marks}}.$$

**Notation.** Event stream  $\{(t_i, k_i)\}$ , marks  $k \in \{1, \dots, K\}$  (single-item,  $K=62$ ). History  $\mathcal{H}_t$ . Hidden width  $H$ . Per-mark intensity  $\lambda_k^*(t)$ , total rate  $\lambda(t) = \sum_k \lambda_k^*(t)$ .

## 1 Backbone: LLH stack $\rightarrow$ embedding $u(t)$

Stacked state-space layers  $\ell = 1, \dots, L$  with diagonal state  $x^{(\ell)}$ , decay  $\delta^{(\ell)} = \text{softplus}(\cdot) + \delta_{\min} > 0$  (optionally input-gated), ZOH-discretized so the state between events evolves  $x_{t'} = \bar{A}x_t + (\bar{A} - I)\bar{B}u^{(\ell-1)}$  with  $\bar{A} = e^{-\delta\Delta t}$ , and an event of mark  $k$  adds an impulse  $\tilde{E}\alpha_k$  ( $\alpha_k$  = mean channel embedding). Position-wise read-out with GELU, residual and LayerNorm gives the (paper-faithful, bounded) embedding  $u(t) := u^{(L)}(t) \in \mathbb{R}^H$ :

$$y^{(\ell)} = C^{(\ell)}x^{(\ell)} + D^{(\ell)}u^{(\ell-1)}, \quad u^{(\ell)} = \text{LayerNorm}(\text{GELU}(y^{(\ell)}) + \text{skip}(u^{(\ell-1)})).$$

This is *identical* to S2P2; both models read the same  $u(t)$ .

## 2 Rate head: gated bound (G1) $\rightarrow \lambda(t)$

A multiplicative gate produces a *bounded* state  $h$ , read out in S2P2’s *Proj & Softplus* form with a learnable positive scale  $s$ :

$$o(t) = \sigma(W_o u(t) + b_o) \in (0, 1)^H, \quad \boxed{h(t) = o(t) \odot \tanh(u(t)) \in (-1, 1)^H}$$

$$\boxed{\lambda(t) = s \cdot \text{softplus}(w_{\text{eff}}^\top h(t) + b)} \quad , \quad s = \text{softplus}(s_{\text{raw}}) > 0, \quad w_{\text{eff}} = w \cdot \min\left(1, \frac{c}{\|w\|_1}\right).$$

The  $\ell_1$  cap enforces  $\|w_{\text{eff}}\|_1 \leq c$ , hence  $|w_{\text{eff}}^\top h| < c$  (since  $|h_i| < 1$ ), giving a closed-form **two-sided bound** — the “Poisson sandwich”:

$$\boxed{\lambda(t) \in (\ell_-, \ell_+), \quad \ell_{\pm} = s \text{softplus}(b \pm c)} \implies \text{Poisson}(\ell_-) \preceq N(t) \preceq \text{Poisson}(\ell_+).$$

The scale  $s$  carries the *magnitude* (mean rate);  $\text{softplus}(b \pm c)$  carries the *multiplicative dynamic range*. Initialise  $b=0$ ,  $s=R_{\text{target}}/\ln 2$  so the baseline rate ( $h \rightarrow 0$ ) is  $\approx R_{\text{target}}$ . **Calibration-free:**  $s, b$  and the MLE  $-\int \lambda$  term set the mean — no EMA, no branching/subcriticality projection.

## 3 Mark head: rate-neutral (decoupled) $\rightarrow p^*(k | t)$

Marks read the *same*  $u$  through an independent softmax MLP and never perturb the total rate:

$$z(t) = W_2 \text{ReLU}(W_1 u(t) + b_1) + b_2 \in \mathbb{R}^K, \quad p^*(k | t) = \text{softmax}(z(t))_k,$$

$$\boxed{\lambda_k^*(t) = \lambda(t) p^*(k | t)} \quad \left( \sum_k \lambda_k^* = \lambda : \text{marks are rate-neutral} \right).$$

## 4 Likelihood (training objective)

Set-MTPP factorization: one ground/total intensity per event time plus a conditional mark term, minus the compensator,

$$\log \mathcal{L} = \sum_i \left[ \log \lambda(t_i) + \log p^*(k_i | t_i) \right] - \int_0^T \lambda(s) ds,$$

with  $\int \lambda$  via the rectangle/quadrature rule on the SSM-evolved state used elsewhere in the codebase. The mark term is the single-item softmax cross-entropy on genuine events.

## 5 Why “SS”: stability properties

1. **No runaway / no death:**  $0 < \ell_- \leq \lambda(t) \leq \ell_+ < \infty$  in closed-loop rollout, by construction — *without* a subcriticality cap.
2. **Self-limiting gain:**  $\partial \lambda / \partial u \rightarrow 0$  as tanh/gate saturate, and  $\text{softplus}' < 1$ .
3. **BIBO state:**  $\delta^{(\ell)} \geq \delta_{\min} > 0 \Rightarrow u(t)$  bounded  $\Rightarrow$  the event  $\rightarrow u \rightarrow \lambda \rightarrow$  event loop is bounded.
4. **Fail-safe baseline:** if the gate closes ( $o \rightarrow 0$ ) then  $h \rightarrow 0$  and  $\lambda \rightarrow s \text{ softplus}(b)$  — free-running reverts to the baseline rate off-distribution.
5. **Single safety knob:** the cap  $c = \|w\|_1$  sets both the dynamic range ( $\approx e^c$  near the soft knee) and the ceiling  $\ell_+$ .

## Implementation map

`ss2p2_decoder.py`: `ground_intensity =  $\lambda$  (G1), mark_score =  $z$ , _w_eff =  $w_{\text{eff}}$ , rate_bounds = ( $\ell_-$ ,  $\ell_+$ ); backbone inherited from S2P2SetDecoder with readout_mode="output". volume_set_mtp.py: the is_ss2p2 branch sets total =  $\lambda$ , marks = softmax( $z$ ), channel =  $\lambda p^*$ . Defaults:  $L=2$ ,  $H=64$ ,  $c=6$ ,  $R_{\text{target}}=3.77$  (Gemini-ETH).`